

AAE5303 Robust Control Technology in Low-Altitude Aerial Vehicle

Post-Lesson Reflection Report

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Submission Requirements

- Length: 800–1200 words | Format: Submit in Github ([Reflection Report](#)) | Individual work [please give a thought before you ask AI to polish your input]
- Use specific examples from YOUR project – avoid generic statements
- Delete all example text (in blue boxes) before submission

Section 1: AI Usage Experience

Describe how you used Cursor (or other AI coding assistants) during the AAE5303 project. Be specific about tasks, frequency, and which features helped you the most.

In the whole project, I used Cursor directly when building Dockerfile. I will use Gemini when analyzing the needs of each job. I will upload the Github link and library of the demo to Gemini to let me analyze the needs of the job. When using homework, I use Gemini almost every day. I upload a lot of course files to him and use the context to help me understand the YAML parameter file in combination with the course content and analyze the results obtained under different parameters to adjust the appropriate parameters. In-line code completion is really easy to use, so I don't have to check the environment variables manually.

Tip: *Be honest about how often you used AI. There is no penalty for using it frequently – what matters is that you demonstrate awareness.*

Section 2: Understanding AI Limitations

Describe a specific situation where the AI produced incorrect, misleading, or hallucinated results. This section is critical – you must provide concrete evidence of AI limitations.

When the ORB-SLAM3 parameter tuning of the AMtown03 data set in the group work, the trajectory tracking failure occurred frequently. AI suggests that I adjust the nFeatures parameters to 3000 or 4000 to extract more dense point clouds and strengthen geometric constraints. However, it did not improve the situation, but made the situation more serious, because the CPU computing power was seriously overloaded, causing the visual tracking thread to be unable to process image frames in time. The terminal began to report an error *Fail to track local map!*, and forced the system to constantly create new maps. I realize that AI's suggestion is based on the ideal assumption of "unlimited computing power". In order to solve this problem, I kept the number of feature points at a moderate level, while lowering the iniThFAST threshold to 15 to capture the features of the low-contrast area, and deliberately reduced the playback speed of rosbag to 0.3x.

Tip: This is the most important section. Showing you can identify when AI is wrong demonstrates critical thinking. Include error logs, screenshots, or config diffs if possible.

Section 3: Engineering Validation

Explain how you validated AI-generated code or solutions to ensure they were correct and reliable.

The official RTK true data is 50Hz, while our monocular camera runs at 10Hz. At the beginning, AI suggested that I write a complex Python script for time stamp interpolation alignment. However, in order to ensure the absolute purity of the data and avoid the introduction of interpolation errors, I wrote an extremely accurate frequency reduction command by myself by verifying the time correspondence, and strictly extracted the data every 5 frames in the true value (*awk 'NR % 5 == 1'*). Subsequently, I used the evo evaluation tool to conduct the final verification test and found that the optimized configuration achieved a trajectory integrity of up to 91.08%, successfully associated 5,646 frames in 6199 frames, obtained a scale correction coefficient of 4.05. This method proves that the 0.3x speed reduction strategy overcomes the hardware bottleneck without sacrificing the rigour and accuracy of the algorithm at all.

Section 4: Problem-Solving Process

Describe a major technical challenge you encountered during the project. Walk through your complete reasoning and problem-solving process.

In this project, I need to try to run the Mono-Inertial/VIO mode. In the data set we use, simply adjusting the parameters cannot greatly solve the problem of system accuracy. Even if the

parameters are modified according to AI or experience, it will lead to a decrease in accuracy. I want to integrate IMU data to improve system accuracy. However, once connected to the IMU data stream, the ORB-SLAM3 system will have catastrophic frame drops, resulting in the completely unuseable output trajectory. At first, I thought it was a parameter calculation error. I spent a lot of time troubleshooting with AI, repeatedly checking the external parameter matrix T_{bc} between the camera and IMU, verifying the noise density parameters of the gyroscope, and checking the time synchronisation logic in the code. AI has also fallen into the blind spot of "code error correction", constantly suggesting that I fine-tune the YAML calibration parameters. It always assumes that the underlying mathematical calculation is wrong. Later, I found that the problem may also be related to the local computing power limitation. Even if the playback speed is reduced to the extreme 0.1x, the data intake still exceeds the processing limit of the CPU. I went back to the initial VO mode to find a solution.

Section 5: Learning Growth

Reflect on how your technical skills and confidence have developed throughout the course.

At the beginning of the semester, I never used Docker or ROS. Although I used AI to assist in the smoke test of homework 1, I did not successfully establish a Dockerfile. But by assignment2, I have successfully been able to establish a Dockerfile to pull the mirror image and configure the ROS workspace. At that time, I was just a user of ready-made code. Today, I have the confidence to go deep into the C++ core source code for modification. For example, I successfully modified the System.cc file in the ORB-SLAM3 underlying library, lifting the original system's restriction on prohibiting monocular cameras from saving TUM tracks.

Section 6: Critical Reflection

Evaluate the overall role of AI in your learning experience. Be honest and balanced.

I think AI is a double-edged sword. On the one hand, AI can improve my learning efficiency, especially when my code foundation is relatively weak. I can quickly get started with relevant homework with the help of AI and explain grammar and content to me. But over-reliance on AI will reduce my own time to think. If you blindly accept the advice given by AI, it may aggravate the problem, causing the problem to accumulate layer by layer, resulting in the complete collapse of the final project. And when AI gives parameter suggestions, it may ignore the computing power limitations of the local environment, resulting in reverse optimisation. Therefore, in the future practice, I must evaluate and accept it in combination with specific hardware constraints and physical rules before accepting the advice of AI.

Section 7: Evidence (Optional but Recommended)

Include supporting evidence to strengthen your reflection. This section can earn bonus credit.

7.1 Code Snippets

Include short, relevant code snippets that illustrate your points. For example, the before/after of a bug fix, or an AI-generated function you had to correct.

To ensure rigorous 10Hz evaluation against the 50Hz ground truth without introducing interpolation errors, I utilized the following precise downsampling script:

```
# Extracting 10Hz ground truth for fair evaluation
awk 'NR % 5 == 1' ground_truth.txt > ground_truth_10hz.txt
```

7.2 Error Logs

Paste relevant error messages or terminal output that illustrate problems you encountered and solved.

```
Fail to track local map!
Fail to track local map!
Fail to track local map!
Creation of new map with id: 4
Stored map with ID: 3
Creation of new map with last KF id: 580
^CShutdown

Saving camera trajectory to CameraTrajectory.txt ...
Segmentation fault (core dumped)
```

7.3 Prompt Examples

Show examples of prompts you gave to Cursor and the responses you received. Highlight cases where the AI was helpful and cases where it was misleading.